Gudla Pushkar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.58
Narayana Junior College	Class XII (Telangana Board of Intermediate Education)	2024	97.00%
Delhi Public School Hyderabad	Class X (Central Board of Secondary Education)	2022	93.60%

SCHOLASTIC ACHIEVEMENTS

Selected to represent **IIT Hyderabad** at the **Inter-IIT Tech Meet 14.0** as part of a **4-member team** tackling the **Game Development** problem statement presented by **GDAI**

JameGam #56 (Submission): Participated as a **solo developer** in the week-long **JameGam #56** game development competition and ranked in the **Top 10** for **Theme Implementation**

POSITIONS OF RESPONSIBILITY

AIML Junior Core, Epoch (Artificial Intelligence & Machine Learning Club of IIT Hyderabad)

Game Development Junior Core, Glitch (Gaming Club of IIT Hyderabad)

SKILLS

Programming Languages: C, C++, Python, JavaScript, SQL, C# (Unity)

Frameworks & Libraries: PyTorch, Scikit-learn, NumPy, Pandas, OpenCV, React, Node.js, Express.js, Socket.IO

Tools & Technologies: Git, GitHub, Linux, Jupyter Notebook, LaTeX, REST APIs, Unity

PROJECTS

Real-Time Bird Species Detection Pipeline (Link)

- Built a **two-stage computer vision pipeline** using **YOLOv8-Nano** for bird detection and **EfficientNet-B1** for fine-grained classification, enabling **real-time recognition of 500 bird species** from images and videos
- Achieved **99.5% mAP@50** with **4.83 ms inference** by integrating **SAHI** for small-object detection and **ByteTrack** with a **Kalman Filter** for robust multi-object tracking across consecutive video frames
- Trained the detection and classification models on **custom** and **Birdsnap** datasets using advanced data augmentation, and deployed the complete pipeline as an interactive **Streamlit** web application for real-time inference

AI Chef (Link)

- Developed a full-stack AI recipe generator using **React**, **Vercel Serverless Functions**, and the **Hugging Face Inference API** to generate personalized recipes from user-provided ingredients
- Built a secure backend with environment variables for LLM inference, implemented **asynchronous API** communication, and rendered AI responses as formatted **Markdown** for improved readability

Sentiment Analysis on IMDB Reviews (Link)

- Compared **16 PyTorch deep learning models** for sentiment classification, including **RNN, LSTM, BiRNN, BiLSTM**, and multiple **attention mechanisms**, using the **HuggingFace IMDB dataset** with custom **PyTorch DataLoaders** and **100-dimensional GloVe embeddings**.
- Achieved **88.65% test accuracy** and a **0.90 F1 score** using an **LSTM with Luong General Attention**, outperforming other evaluated architectures through comprehensive experimental comparison

Evaluating XAI Robustness via Adversarial Attacks | *AI for Humanity Course Project* | (Link)

- Implemented **Scaffolding** and **Gradient-Based adversarial attacks** to evaluate the robustness of **LIME** and **SHAP**, demonstrating how post-hoc explanations can be manipulated while preserving **model fairness**, **prediction quality**, and **classification performance on real-world data**
- Built an end-to-end evaluation pipeline using **Fairlearn**, **Random Forests**, **MLPs**, and an **OOD detector** on the **Adult Income** and **COMPAS** datasets, measuring **attack effectiveness**, **fairness metrics**, **explanation robustness**, and the effectiveness of proposed **defense mechanisms against explanation manipulation**

Dual-World Puzzle Platformer (Inter-IIT Tech Meet 14.0 Submission) (Link)

- Developed a **Unity-based game** combining **3D exploration** and a **2D puzzle platformer** with **interconnected progression**, seamless scene transitions, and persistent game state
- Implemented **player movement**, **color-switching puzzles**, **dash**, **wall jump**, **checkpoints**, **moving platforms**, and interactive environmental mechanics using **C#**

RELEVANT COURSES

Artificial Intelligence: Introduction to Modern AI, Intro to Classical AI, Programming for AI, Foundations of Machine Learning, Deep Learning, AI for Humanity

Computer Science: Introduction to Programming, Data Structures & Applications, Algorithms, Operating Systems-1, Database Management Systems-I

Mathematics: Discrete Mathematics, Matrix Theory, Calculus I & II, Vector Calculus, Differential Equations, Probability & Random Variables, Introduction to Statistics, Optimization I & II, Numerical Analysis, Linear Systems & Signal Processing

EXTRACURRICULAR

Assisted in organizing the **Glitch Spring Camp** and the **HP OMEN × Glitch** collaboration event

Contributed to an **Epoch-led machine learning resource guide**, curating tutorials, courses, and reference material to make ML learning more accessible